Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **Python test automation**, **unit and integration test suites**, and Selenium **GUI testing**, turning requirements into executable test steps with clear test reporting and **cross-functional collaboration**.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Implemented **data validation** and reconciliation checks in Python FastAPI integration pipelines across PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs through CI/CD releases at 99%+ production reliability.
- Restructured a fault-tolerant async LLM orchestration system with explicit job-state transitions and **failure isolation**, processing 1K+ jobs and tripling throughput under concurrent load without cascading task failures.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Built a **Python** Flask grading backend handling text and image answers through model-specific execution paths, with Claude Vision fallback for OCR timeouts, serving 150+ instructors and 10K+ daily requests.
- Shipped CI/CD-driven releases with **GitHub Actions** and logged token usage and estimated cost for every model call, maintaining stable API responses under peak load while cutting feedback turnaround 60%.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles*

- Built **Python data processing** pipelines converting 100K+ autonomous-driving trajectory logs into model-ready features, tracking experiments with Hydra configurations and Weights & Biases for reproducible runs and results analysis.
- Evaluated a Decision Transformer planner in **closed-loop simulation** against Wayformer-style and MotionLM-style baselines, analyzing off-road and lane-following behavior and improving long-horizon trajectory planning 18% and closed-loop stability 15%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Automated nightly **Jenkins** test runs that pulled the latest commits, built executables, installed them on dedicated VMs, and executed **integration tests** across end-to-end synchronization flows on Windows, Linux, and macOS.
- Wrote 20+ unit and integration test suites with **PyTest** and Mockito, mocking isolated components and generating CSV reports, coverage summaries, and failure logs while collaborating with Microsoft and GitHub stakeholders.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Ran unit, integration, and **end-to-end tests** with Jest, Mocha, and **Selenium** in Jenkins pipelines before Docker packaging, achieving 95%+ successful build validation and gating production behind manual approval steps.
- Integrated Snyk and OWASP Dependency-Check scans with Sentry error tracking, Azure Monitor metrics, and Kibana logs for a MERN resource platform synchronizing 2M+ records across locations and release environments.

Skills

Testing & Verification: Test Automation, Test Case Design, Unit Testing, Integration Testing, End-to-End Testing, GUI Testing, PyTest, Mockito, Jest, Mocha, Selenium, Code Coverage, Debugging, Root-Cause Analysis
Languages & Libraries: Python, C++, Java, JavaScript, TypeScript, SQL, Bash, Shell, PyTorch, Hydra
Tools & Version Control: Git, GitHub, JIRA, Jenkins, GitHub Actions, CI/CD, Docker, Linux, Sentry, Snyk, Weights & Biases
Practices: Results Analysis, Test Reporting, Technical Documentation, Agile/Scrum, SDLC, Cross-Functional Collaboration

Projects

CPU-scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*

- **Tested** FCFS, SJF, SRT, Round Robin, priority, and MLQ scheduling for correctness across preemptive, non-preemptive, and edge-case scenarios over 1K+ simulated process combinations, reducing debugging time 20%.

HMIS - Hospital Management Information System Live | *TypeScript, MERN, ABDM APIs, Twilio, Sentry*

- Built a **healthcare** platform with consent-based patient-record access through ABDM APIs, Sentry error monitoring, and Twilio reminders, supporting 100+ daily appointments and reducing administrative workload 30% for staff.

CaptionCraft - Image Caption Generator Live | *Python, PyTorch, Transformers, Flask*

- Benchmarked fine-tuned BLIP, GIT, ViT-GPT-2, and a custom model in **Python** on a 10K-image, 100K-caption dataset, comparing results to raise caption accuracy from 81% to 96% for inference APIs.